

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

Product Name

Electrolyte

Product Code

**S60201**

Address

ThermoFisher Scientific Australia Pty Ltd  
 5 Caribbean Drive, Scoresby  
 VICTORIA 3179, Australia

Emergency Tel.

**CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

Telephone / Fax Numbers

Tel: 1300 735 292  
 Fax: 1800 067 639

E-mail address

ANZinfo@thermofisher.com

Recommended Use

Laboratory chemicals.

Uses advised against

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 3

#### Health hazards

Acute Oral Toxicity  
 Skin Corrosion/Irritation  
 Serious Eye Damage/Eye Irritation  
 Specific target organ toxicity - (repeated exposure)

Category 4  
 Category 1 A  
 Category 1  
 Category 1

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Exclamation Mark



Health Hazard



Corrosion

**Signal Word****Danger****Hazard Statements**

H226 - Flammable liquid and vapor

H302 - Harmful if swallowed

H314 - Causes severe skin burns and eye damage

H372 - Causes damage to organs through prolonged or repeated exposure

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P240 - Ground and bond container and receiving equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P331 - Do NOT induce vomiting

P363 - Wash contaminated clothing before reuse

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

No information available

**Section 3 - Composition and Information on Ingredients**

| Component                           | CAS No      | Weight % |
|-------------------------------------|-------------|----------|
| Diethyl carbonate                   | 105-58-8    | 54.21    |
| Ethylene carbonate                  | 96-49-1     | 26.24    |
| Phosphate(1-), hexafluoro-, lithium | 21324-40-3  | 12.61    |
| 4-Fluoro-1,3-dioxolan-2-one         | 114435-02-8 | 5.95     |
| Vinylene carbonate                  | 872-36-6    | 0.99     |

**Section 4 - First Aid Measures****Inhalation**

If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.

**Ingestion**

Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

|                                            |                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                             |
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                    |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                     |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                 |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting; Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                                                                               |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Carbon oxides, Lithium oxide, Hydrogen fluoride, Oxides of phosphorus.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

**Conditions for Safe Storage, Including any Incompatibilities**

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component                           | Australia                  | New Zealand WEL | ACGIH TLV                  | The United Kingdom | Germany                                                                                                         |
|-------------------------------------|----------------------------|-----------------|----------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------|
| Phosphate(1-), hexafluoro-, lithium | TWA: 2.5 mg/m <sup>3</sup> |                 | TWA: 2.5 mg/m <sup>3</sup> |                    | TWA: 1 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 4<br>TWA: 1 mg/m <sup>3</sup> (8 Stunden). MAK Haut |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Neoprene       | recommendations   |                 |                 |                       |
| Natural rubber |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b> | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>   | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b> | Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                                                                                  |
| <b>Recommended half mask:-</b>  | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                         |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                              | Clear                    |                                          |
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flash Point</b>                             | 25 °C / 77 °F            | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.24                     |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>                        | No information available |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Diethyl carbonate                              | 1.33                     |                                          |
| Ethylene carbonate                             | 0.11                     |                                          |
| 4-Fluoro-1,3-dioxolan-2-one                    | -0.435                   |                                          |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Explosive Properties</b>                    | No information available | explosive air/vapour mixtures possible   |
| <b>Oxidizing Properties</b>                    | No information available |                                          |

### Other information

## Section 10 - Stability and Reactivity

|                                  |                                                                        |
|----------------------------------|------------------------------------------------------------------------|
| Reactivity                       | None known, based on information available                             |
| Stability                        | Stable under normal conditions.                                        |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition.      |
| Incompatible Materials           | None known.                                                            |
| Hazardous Decomposition Products | Carbon oxides. Lithium oxide. Hydrogen fluoride. Oxides of phosphorus. |
| Hazardous Polymerization         | No information available.                                              |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

#### Toxicology data for the components

| Component                           | LD50 Oral               | LD50 Dermal                   | LC50 Inhalation                          |
|-------------------------------------|-------------------------|-------------------------------|------------------------------------------|
| Diethyl carbonate                   | >15000 mg/kg (Rat)      |                               | >19.5 mg/L/4h (Rat)                      |
| Ethylene carbonate                  | LD50 = 10 g/kg ( Rat )  | LD50 > 26420 mg/kg ( Rabbit ) | LC50 > 730 mg/m <sup>3</sup> ( Rat ) 8 h |
| Phosphate(1-), hexafluoro-, lithium | > 50 - 300 mg/kg (Rat)  |                               |                                          |
| 4-Fluoro-1,3-dioxolan-2-one         | LD50 = ca. 500 mg/kg bw | LD50 > 2000 mg/kg ( Rat )     |                                          |
| Vinylene carbonate                  | >300-< 500 mg/kg (Rat)  | LD50 200 - 2000 mg/kg ( Rat ) |                                          |

(b) skin corrosion/irritation; Category 1 A

(c) serious eye damage/irritation; Category 1

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 1

**Target Organs** No information available.

**(j) aspiration hazard;** No data available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## Section 12 - Ecological Information

**Ecotoxicity effects** May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component                           | Freshwater Fish                                      | Water Flea            | Freshwater Algae    | Microtox |
|-------------------------------------|------------------------------------------------------|-----------------------|---------------------|----------|
| Diethyl carbonate                   | Leuciscus idus: LC50 >500 mg/L/48H                   |                       | EC50 >10000 mg/L/3H |          |
| Ethylene carbonate                  | LC50: > 100 mg/L, 96h (Oncorhynchus mykiss)          |                       |                     |          |
| Phosphate(1-), hexafluoro-, lithium |                                                      | EC50 > 100 mg/l (48h) |                     |          |
| Vinylene carbonate                  | LC50: = 2.4 mg/L, 96h flow-through (Cyprinus carpio) |                       |                     |          |

**Persistence and Degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary

**Persistence** May persist.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.  
**Bioaccumulative Potential** Product has a high potential to bioconcentrate

| Component                   | log Pow | Bioconcentration factor (BCF) |
|-----------------------------|---------|-------------------------------|
| Diethyl carbonate           | 1.33    | No data available             |
| Ethylene carbonate          | 0.11    | No data available             |
| 4-Fluoro-1,3-dioxolan-2-one | -0.435  | No data available             |

**Mobility** No information available.  
**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information** Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

## Section 14 - Transport Information

### IMDG/IMO

**UN-No** UN2924  
**Proper Shipping Name** FLAMMABLE LIQUID, CORROSIVE, N.O.S.  
**Technical Shipping Name** Diethyl carbonate/vinylene carbonate  
**Hazard Class** 3  
**Subsidiary Hazard Class** 8  
**Packing Group** III

**ADG**

**UN-No** UN2924  
**Proper Shipping Name** FLAMMABLE LIQUID, CORROSIVE, N.O.S.  
**Technical Shipping Name** Diethyl carbonate/vinylene carbonate  
**Hazard Class** 3  
**Subsidiary Hazard Class** 8  
**Packing Group** III

| Component                               | Hazchem Code |
|-----------------------------------------|--------------|
| Diethyl carbonate<br>105-58-8 ( 54.21 ) | 3Y           |

**IATA**

**UN-No** UN2924  
**Proper Shipping Name** FLAMMABLE LIQUID, CORROSIVE, N.O.S.  
**Technical Shipping Name** Diethyl carbonate/vinylene carbonate  
**Hazard Class** 3  
**Subsidiary Hazard Class** 8  
**Packing Group** III

**Environmental hazards** No hazards identified  
**Special Precautions** No special precautions required  
**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                                           | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phosphate(1-), hexafluoro-, lithium -<br>21324-40-3 | Schedule 2 listed<br>Schedule 3 listed<br>Schedule 4 listed - in preparations for human use except when included in or expressly excluded from<br>Schedule 2 or 3<br>Schedule 5 listed - in preparations except: in preparations for human use, or in preparations containing<br><=15 mg/kg of Fluoride ion; as Fluoride ion<br>Schedule 6 listed - except: when included in Schedule 5, in preparations for human use, or in<br>preparations containing <=15 mg/kg of Fluoride ion |

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component | Australian Industrial | Additional information |
|-----------|-----------------------|------------------------|
|-----------|-----------------------|------------------------|



|                                                  | Chemicals Introduction Scheme (AICIS) |   |
|--------------------------------------------------|---------------------------------------|---|
| Diethyl carbonate - 105-58-8                     | Present                               | - |
| Ethylene carbonate - 96-49-1                     | Present                               | - |
| Phosphate(1-), hexafluoro-, lithium - 21324-40-3 | Present                               | - |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component                           | AICS | NZIoC | EINECS    | ELINCS    | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL        |
|-------------------------------------|------|-------|-----------|-----------|------|-----|------|-------|------|------|-------|-------------|
| Diethyl carbonate                   | X    | X     | 203-311-1 | -         | X    | X   | -    | X     | X    | X    | X     | KE-04706    |
| Ethylene carbonate                  | X    | X     | 202-510-0 | -         | X    | X   | -    | X     | X    | X    | X     | KE-12028    |
| Phosphate(1-), hexafluoro-, lithium | X    | -     | 244-334-7 | -         | X    | -   | X    | X     | X    | X    | X     | KE-22564    |
| 4-Fluoro-1,3-dioxolan-2-one         | -    | -     | -         | 483-360-5 | X    | -   | X    | -     | X    | X    | -     | 2004-3-2827 |
| Vinylene carbonate                  | -    | -     | 212-825-5 | -         | X    | -   | X    | -     | X    | X    | X     | 2001-3-1927 |

**Legend:** X - Listed. '-' - Not Listed. PMN - Indicates a commenced PMN substance. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component | CAS No | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|--------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
|           |        |          |                                            |                                                                                           |                                                                                          |

|                                     |             |                |                |                |                |
|-------------------------------------|-------------|----------------|----------------|----------------|----------------|
| Diethyl carbonate                   | 105-58-8    | Not applicable | Not applicable | Not applicable | Not applicable |
| Ethylene carbonate                  | 96-49-1     | Listed         | Not applicable | Not applicable | Not applicable |
| Phosphate(1-), hexafluoro-, lithium | 21324-40-3  | Not applicable | Not applicable | Not applicable | Not applicable |
| 4-Fluoro-1,3-dioxolan-2-one         | 114435-02-8 | Not applicable | Not applicable | Not applicable | Not applicable |
| Vinylene carbonate                  | 872-36-6    | Listed         | Not applicable | Not applicable | Not applicable |

Authorisation/Restrictions according to EU REACH

Not applicable

## Section 16 - Other Information

### Legend

AICS - Australian Inventory of Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

IECSC - Chinese Inventory of Existing Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

NZS 5433:2012 - Transport of Dangerous Goods on Land

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

WEL - Workplace Exposure Limit

DNEL - Derived No Effect Level

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

VOC - (Volatile Organic Compound)

NZIoC - New Zealand Inventory of Chemicals

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

CAS - Chemical Abstracts Service

ACGIH - American Conference of Governmental Industrial Hygienists Predicted No Effect Concentration (PNEC)

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

LC50 - Lethal Concentration 50%

ATE - Acute Toxicity Estimate

RPE - Respiratory Protective Equipment

NOEC - No Observed Effect Concentration

BCF - Bioconcentration factor

PBT - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

|                       |                       |
|-----------------------|-----------------------|
| Physical hazards      | On basis of test data |
| Health Hazards        | Calculation method    |
| Environmental hazards | Calculation method    |

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Revision Date 26-Jan-2023

Revision Summary Initial Release.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,

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transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**